

Mikroelektronische Schaltungen und Systeme

Single Stage Amplifiers

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Common-Source Amplifier

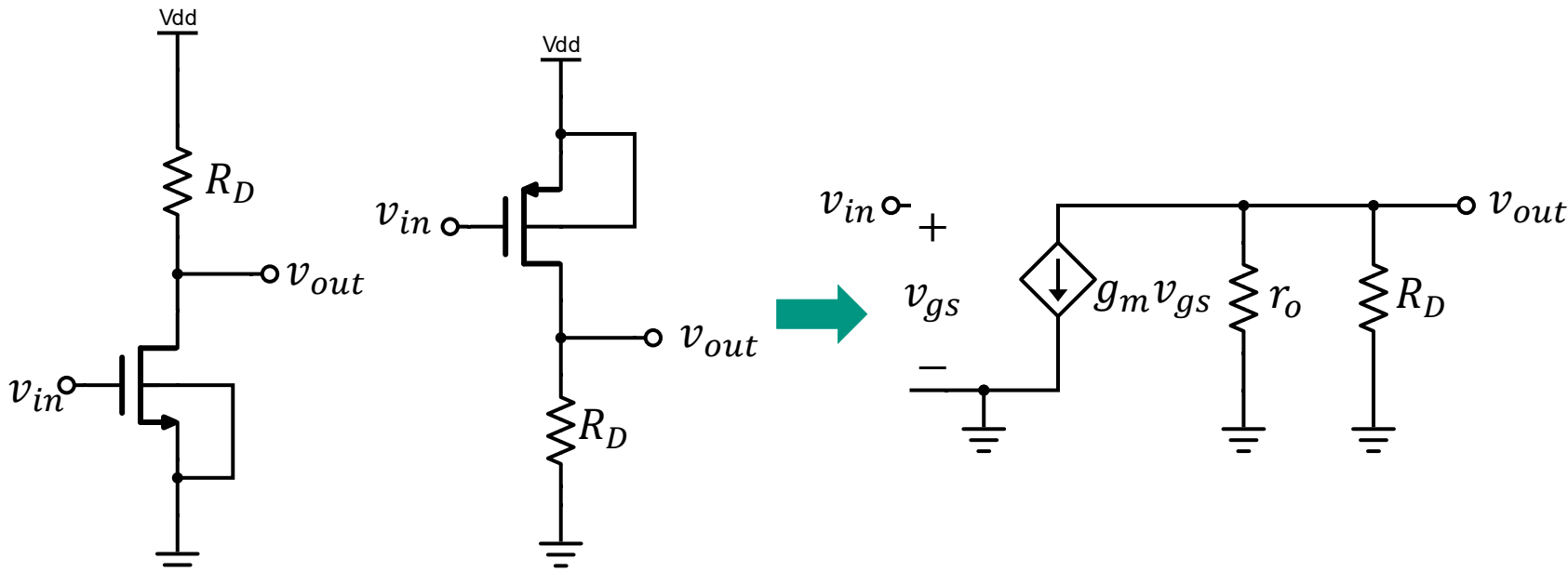
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Letter Convention

- DC Signals (Upper case): V_{GS}, V_B, I_D
- AC Signals (Lower case): v_{gs}, v_x, i_x

Common-Source Stage with Resistive Load

- The common-source amplifier is the most fundamental amplifier block, which was introduced in the ES course.



■ CS NMOS

■ CS PMOS

■ Small-signal model

- The midband gain:

$$A_v = \frac{v_{out}}{v_{in}} = -g_m(r_o \parallel R_D) \approx -g_m R_D$$

- Input & output resistances:

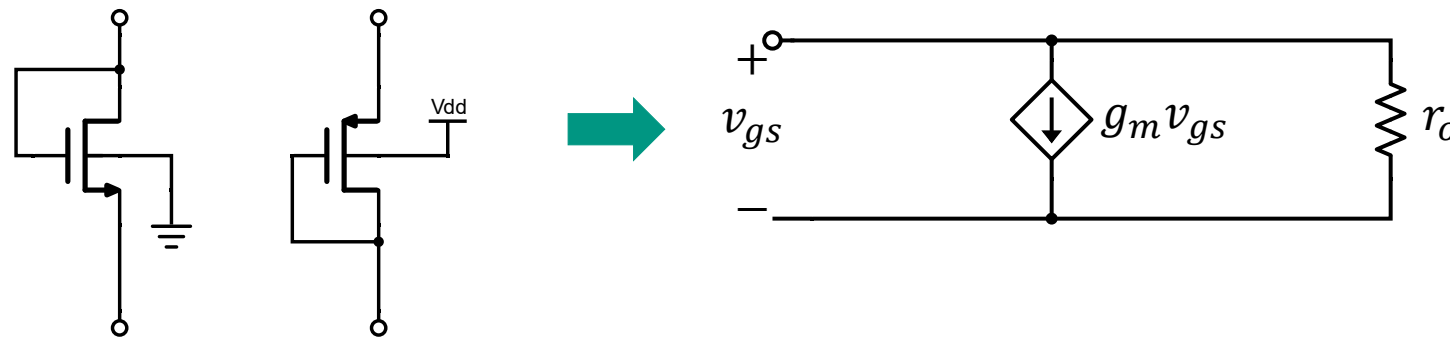
$$R_{in} \rightarrow \infty$$

$$R_{out} = (r_o \parallel R_D)$$

CS Stage with Diode-Connected Load

Diode Connected Device

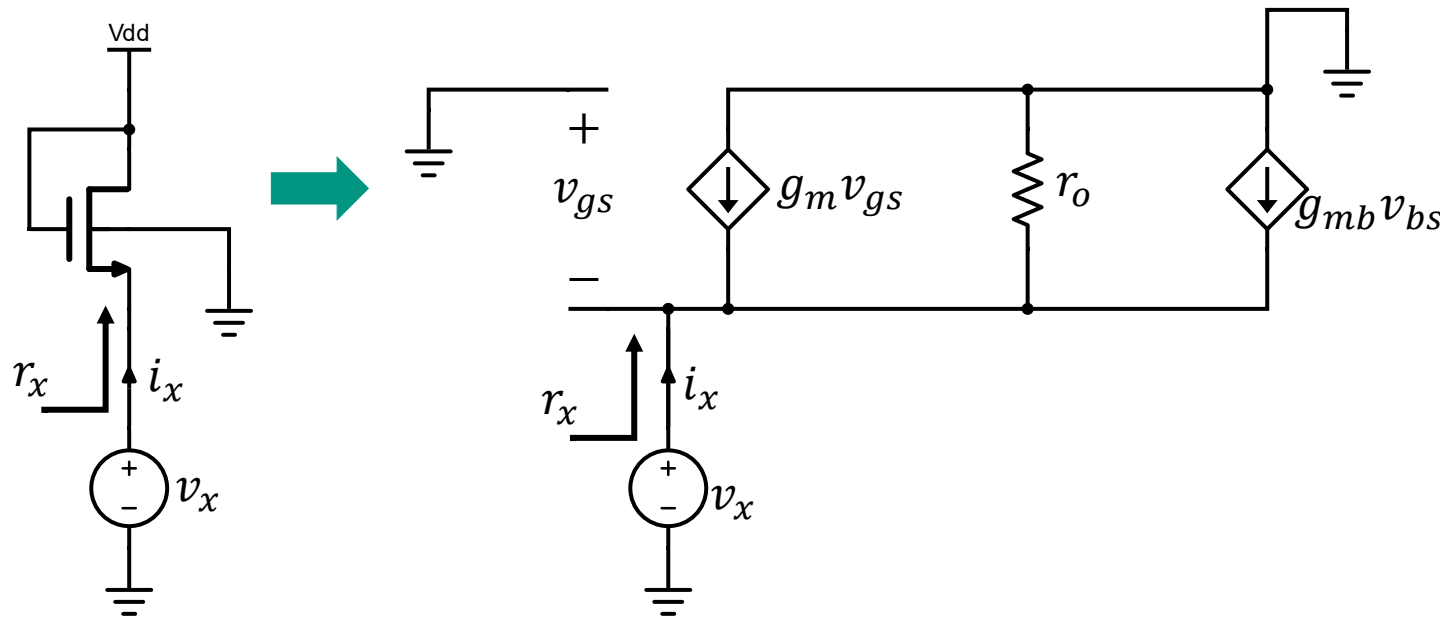
- In CMOS technologies, passive devices occupy large area, and it is challenging to fabricate resistors with high precision, as their resistance can vary significantly.
- To replace these imprecise devices, MOS transistors are often used as a load.
- A MOSFET with its gate and drain shorted can operate as a small-signal resistor. When a voltage is applied from drain to source for NMOS (from source to drain for PMOS), this gate–drain connection forward-biases the channel, which is why it is called a 'diode-connected' MOSFET.



CS Stage with Diode-Connected Load

Analysis of Diode Connected Device

- To use the diode-connected MOSFET as a load, let's calculate the equivalent small-signal resistance (R_x) seen from the source terminal.



$$\begin{aligned}v_{gs} &= -v_x, v_{bs} = -v_x \\i_x &= -g_m v_{gs} + \frac{v_x}{r_o} - g_{mb} v_{bs} \\i_x &= g_m v_x + \frac{v_x}{r_o} + g_{mb} v_x \\r_x = \frac{v_x}{i_x} &= \frac{1}{g_m + \frac{1}{r_o} + g_{mb}} = \frac{1}{g_m} \parallel r_o \parallel \frac{1}{g_{mb}}\end{aligned}$$

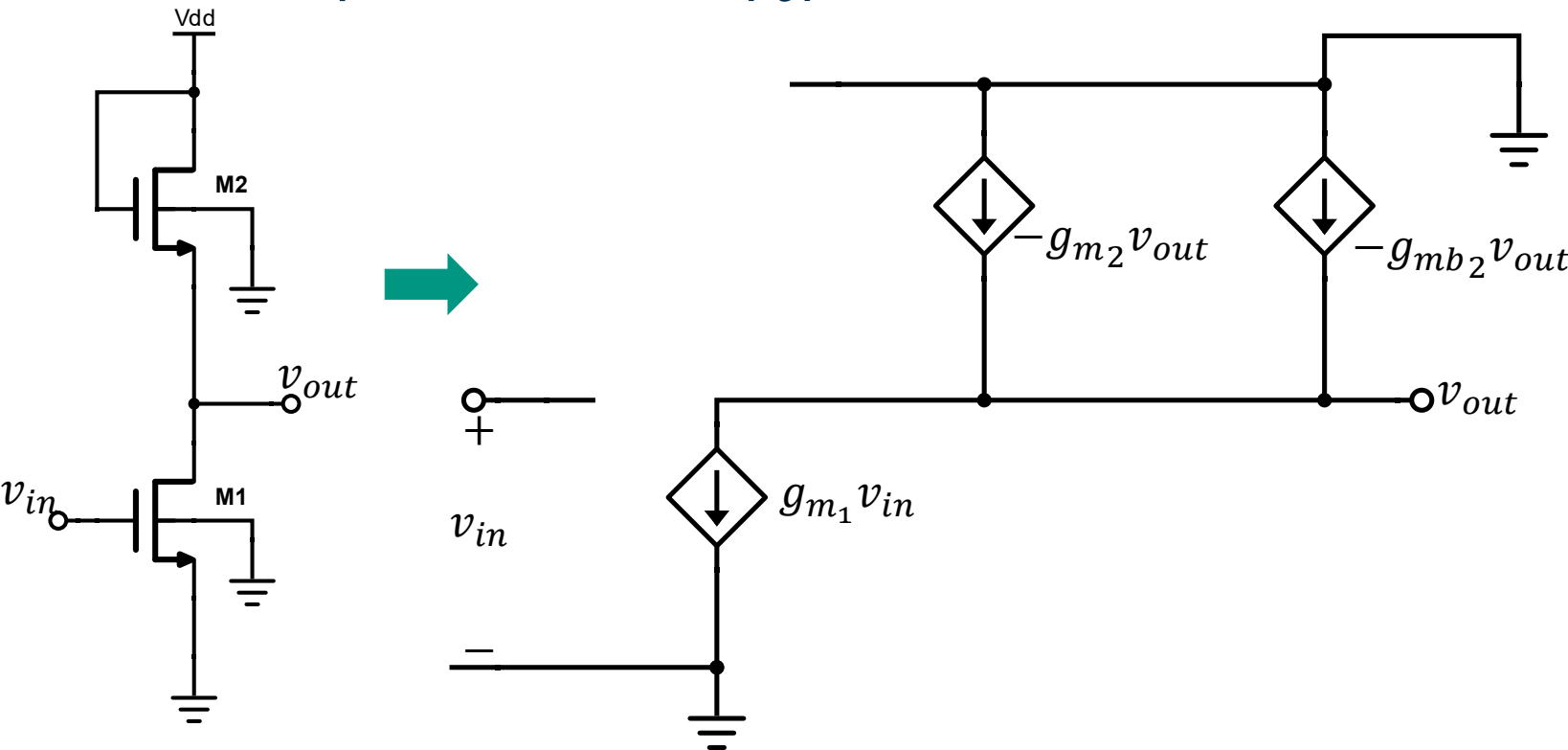
$$\text{Considering } r_o \gg \frac{1}{g_m}$$

$$\boxed{r_x \approx \frac{1}{g_m + g_{mb}}}$$

CS Stage with Diode-Connected Load

Voltage Gain Expression

- The output resistance (r_o) of both transistors are omitted in small-signal model.



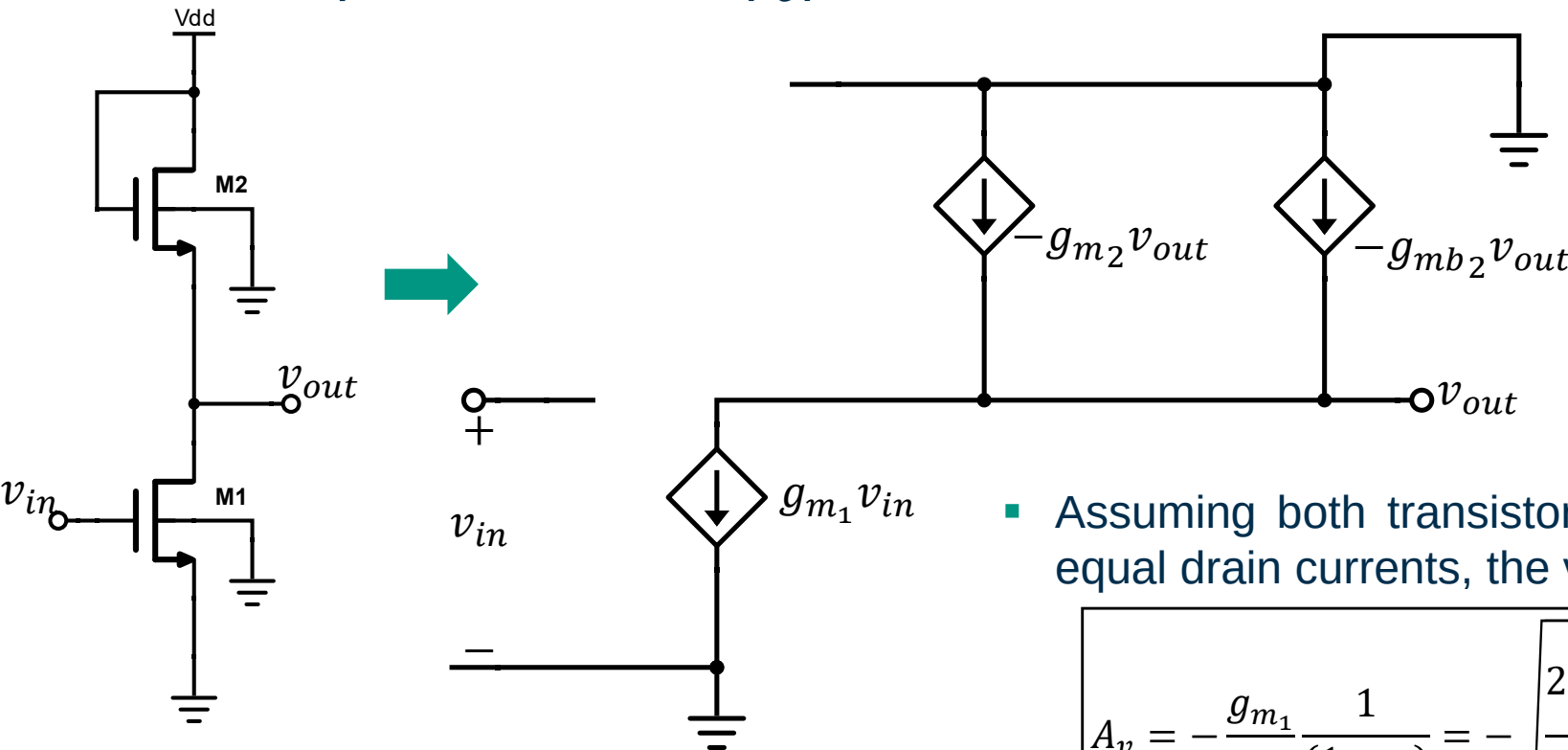
$$g_{m1}v_{in} = -v_{out}(g_{m2} + g_{mb2})$$
$$A_v = \frac{v_{out}}{v_{in}} = -\frac{g_{m1}}{g_{m2} + g_{mb2}} = -g_{m1}r_{x2}$$

(Recall: $\frac{g_m}{g_{mb}} = \frac{1}{\eta}$)

CS Stage with Diode-Connected Load

Voltage Gain Expression

- The output resistance (r_o) of both transistors are omitted in small-signal model.



$$g_{m_1} v_{in} = -v_{out} (g_{m_2} + g_{mb_2})$$

$$A_v = \frac{v_{out}}{v_{in}} = -\frac{g_{m_1}}{g_{m_2} + g_{mb_2}} = -g_{m_1} r_{x_2}$$

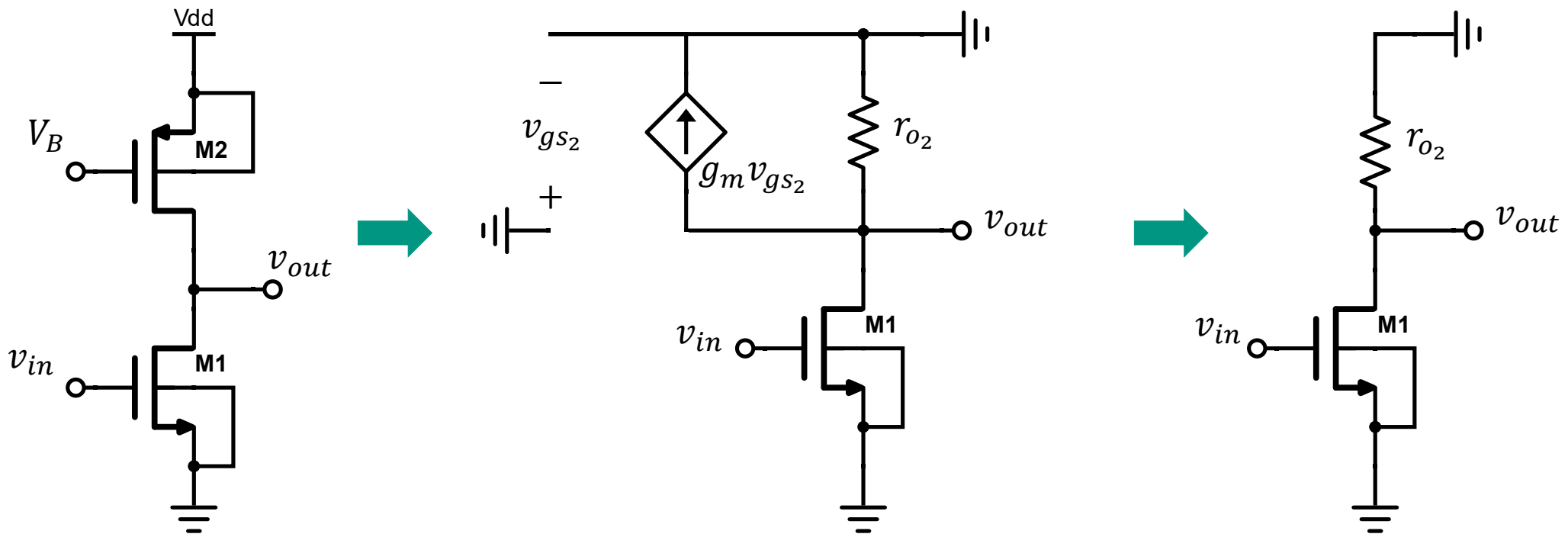
(Recall: $\frac{g_m}{g_{mb}} = \frac{1}{\eta}$)

- Assuming both transistors have the same $\mu_n C_{ox}$ and operate at equal drain currents, the voltage gain can be expressed as:

$$A_v = -\frac{g_{m_1}}{g_{m_2}} \frac{1}{(1 + \eta)} = -\frac{2I_D \mu_n C_{ox} \left(\frac{W}{L}\right)_1}{2I_D \mu_n C_{ox} \left(\frac{W}{L}\right)_2} \frac{1}{(1 + \eta)} = -\frac{\left(\frac{W}{L}\right)_1}{\left(\frac{W}{L}\right)_2} \frac{1}{(1 + \eta)}$$

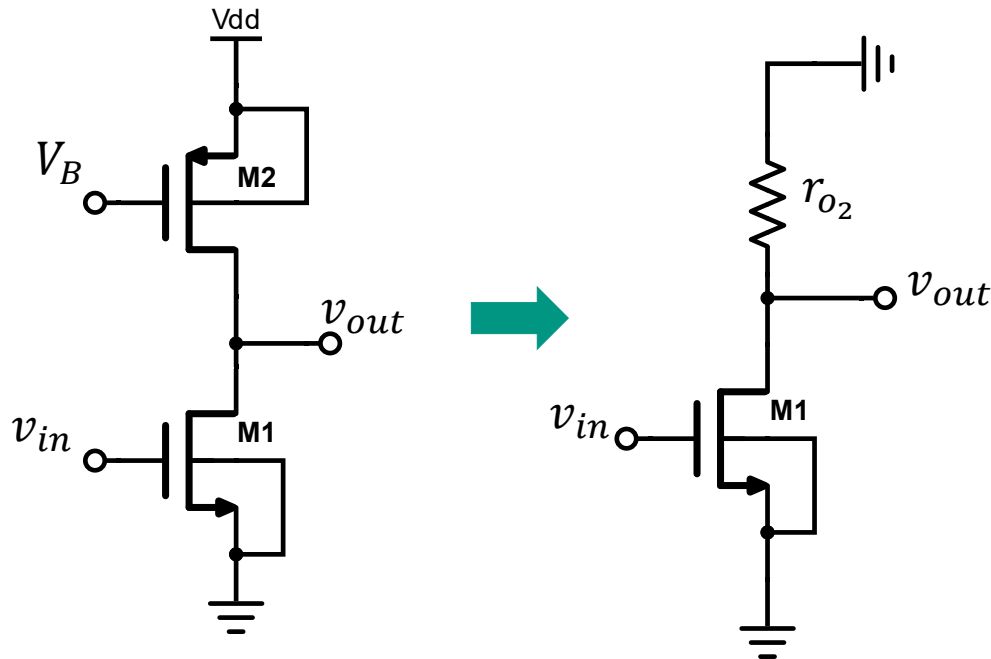
CS Stage with Current-Source Load

- The current-source load of a CS stage can be implemented using a PMOS transistor. Both transistors are biased in saturation region.



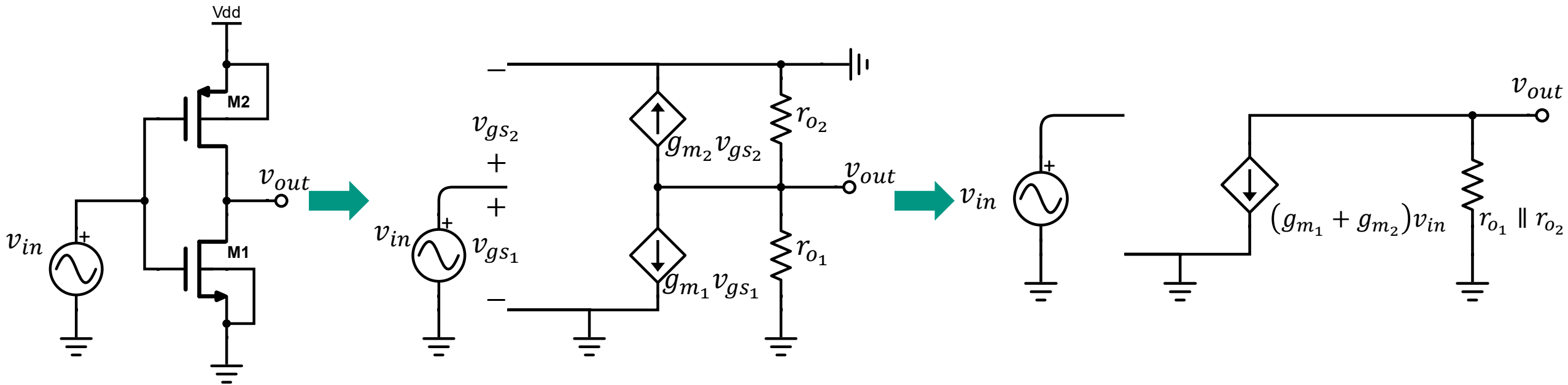
CS Stage with Current-Source Load

- A current-source load provides higher output resistance and is less dependent on voltage drop than a resistive load, The minimum voltage level to keep $|V_{DS}|$ in the saturation ($V_{D,sat}$ or V_{ov}) of the load can be reduced to less than a hundred millivolts, thereby allowing greater voltage headroom.



$$A_v = \frac{v_{out}}{v_{in}} = -g_{m1}(r_{o1} \parallel r_{o2})$$

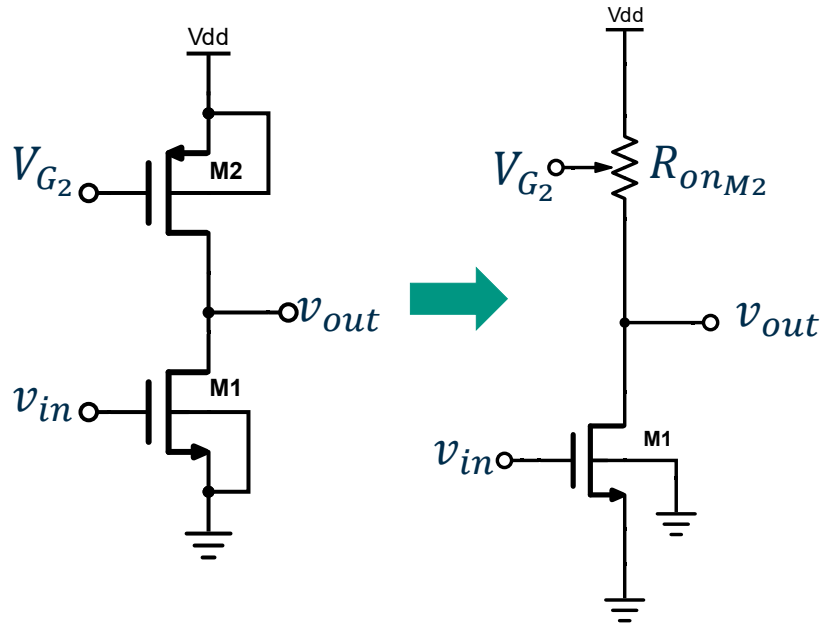
CS Stage with Active Load – CMOS Inverter Amplifier



$$A_v = -(g_{m1} + g_{m2})(r_{o1} \parallel r_{o2})$$

CS Stage with Triode Load

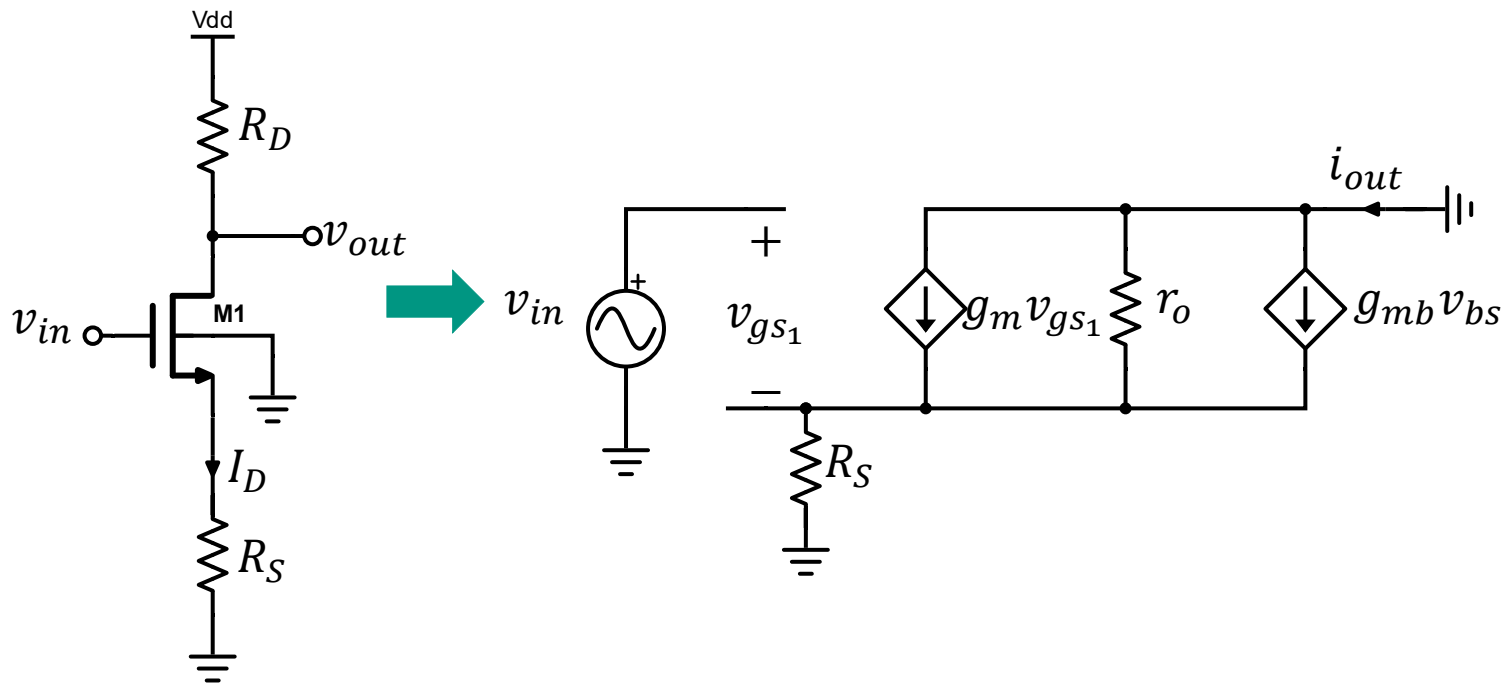
- The load transistor can be operated in triode region. In this way, the load will operate like a variable resistor controlled by the load transistor's gate voltage (V_{G_2}).



$$R_{onM2} = \frac{1}{\mu_p C_{ox} \left(\frac{W}{L}\right)_2 (V_{DD} - V_{G_2} - |V_{THP}|)}$$
$$A_v = -g_{m1} R_{onM2}$$

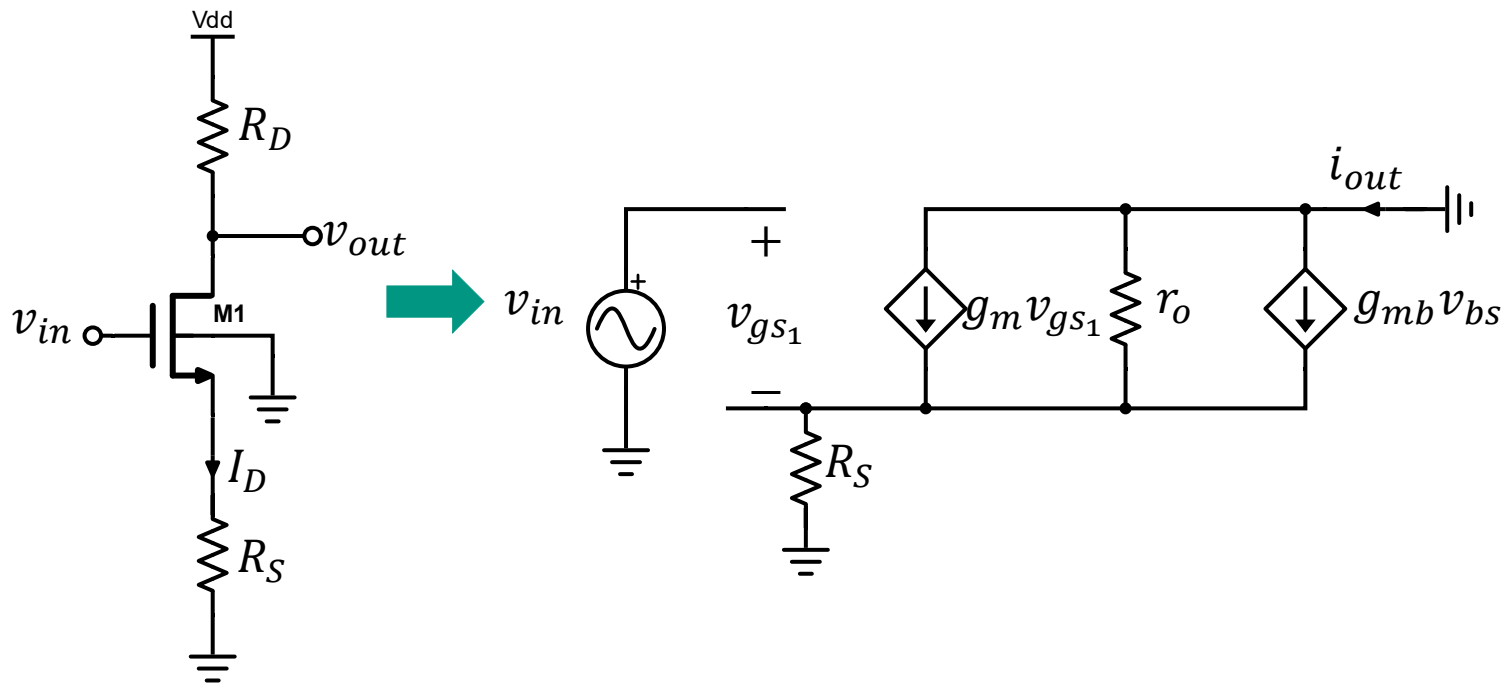
CS Stage with Source Degeneration

- Source degeneration introduces negative feedback by adding a resistor to the source terminal. This technique improves the transistor's linearity, but with the trade-off of lowering its effective transconductance and overall gain.



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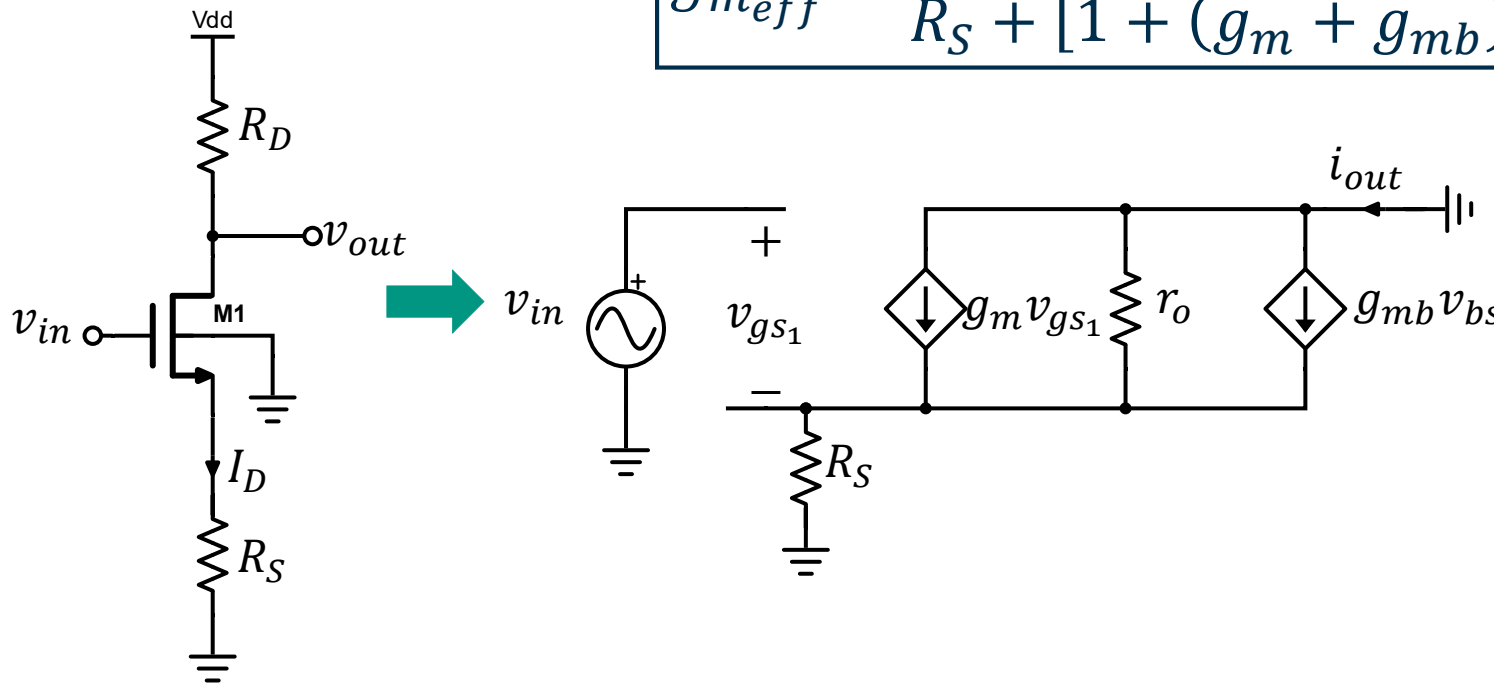
- The simplified effective transconductance:

$$g_{m_{eff}} = \frac{g_m}{1 + g_m R_S}$$

CS Stage with Source Degeneration

- The source degeneration transconductance in the presence of body effect and channel-length modulation (R_D excluded):

$$g_{m_{eff}} = \frac{g_m r_o}{R_S + [1 + (g_m + g_{mb})R_S]r_o}$$



CS Stage with Source Degeneration

Output Resistance of a Source Degenerated CS Stage

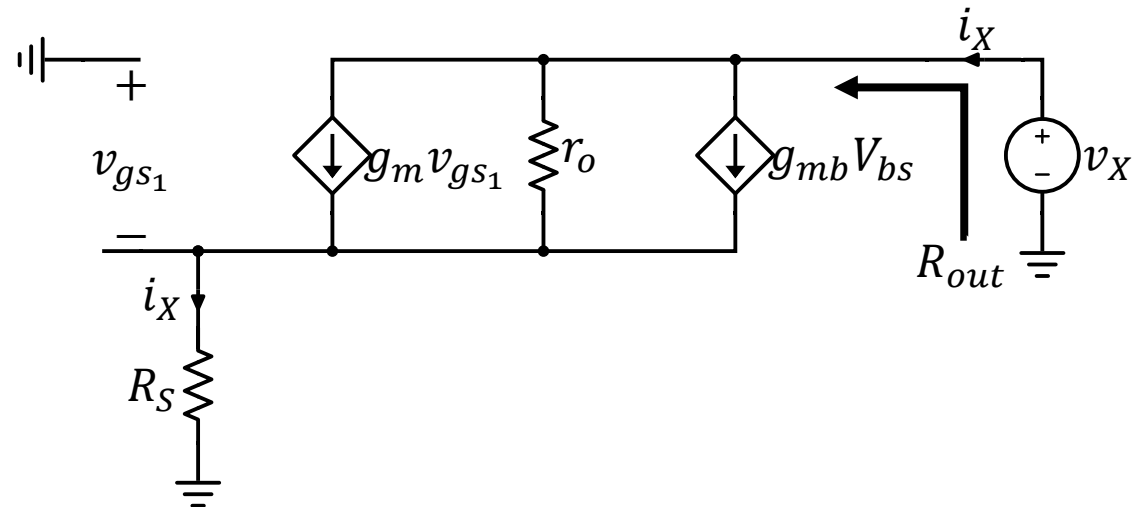
- The output resistance of the source degenerated CS Stage can be derived as:

$$R_{out} \approx [1 + (g_m + g_{mb})R_S]r_o$$

- By ignoring the body effect, simplifying to:

$$R_{out} \approx (1 + g_m R_S)r_o$$

- Thus, the output resistance is boosted by $(1 + g_m R_S)$.



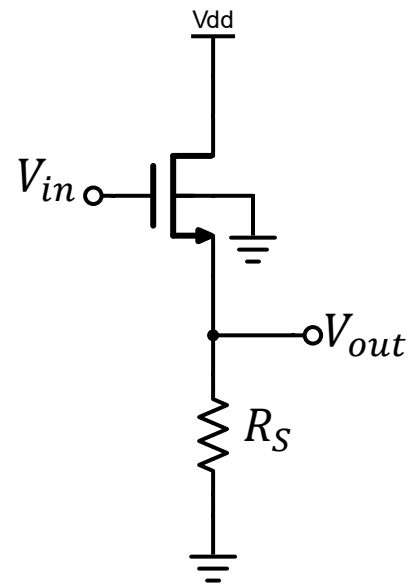


Source Follower

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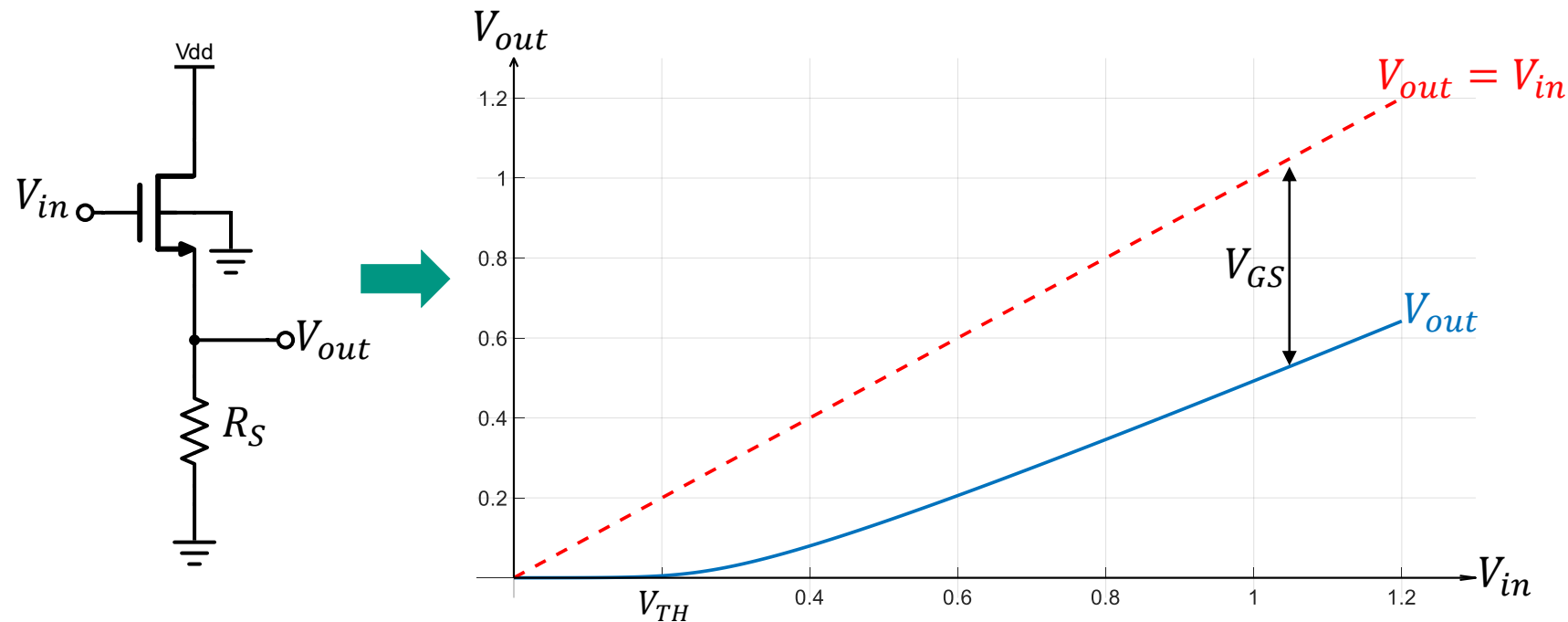
Source Follower (Common-Drain) Stage

- The standard gain formula of the CS Stage ($|A_v| = g_m R_{out}$) indicates that large load impedance results in higher gain. However, if the load impedance is low, the gain will drop. To drive low-impedance loads without losing signal amplitude, a source follower stage is used. For this reason, it is also known as a **buffer stage**.



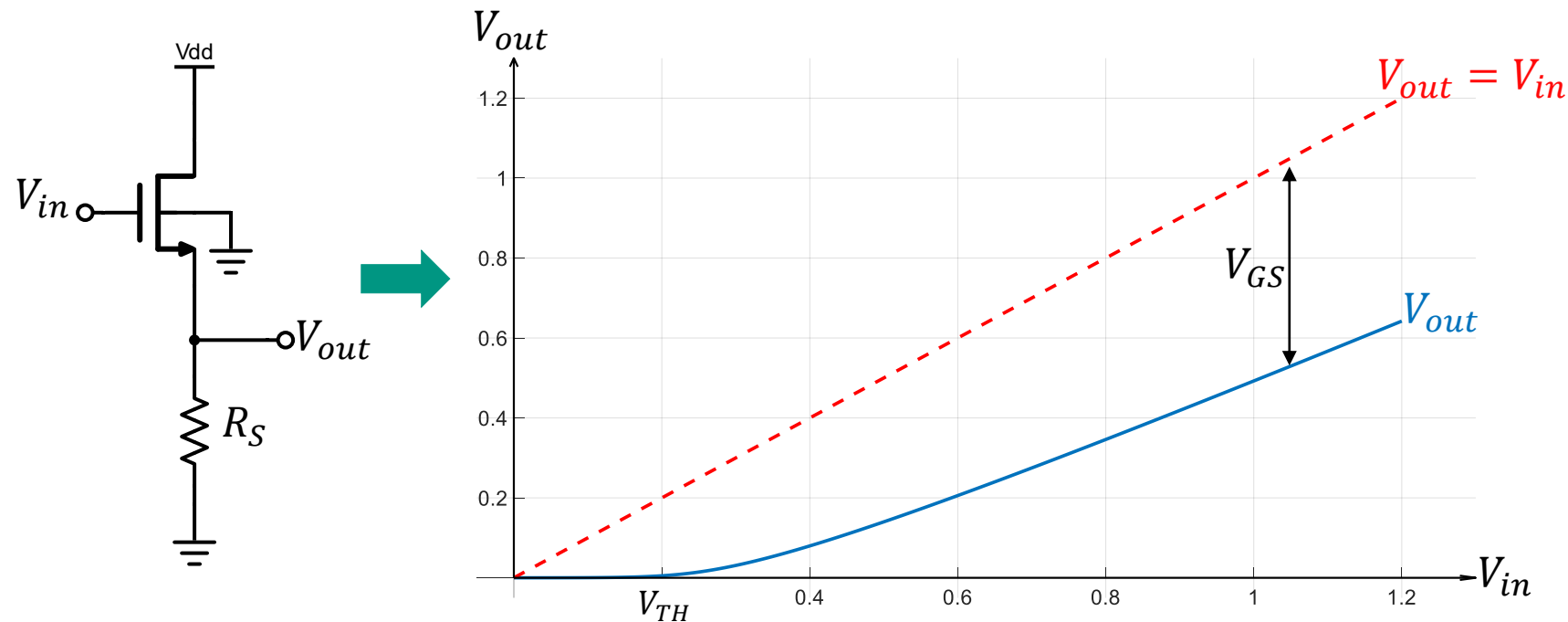
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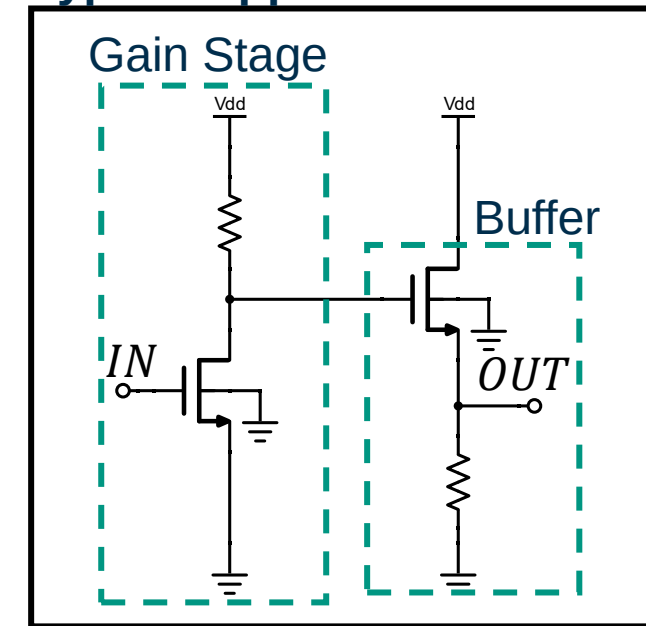


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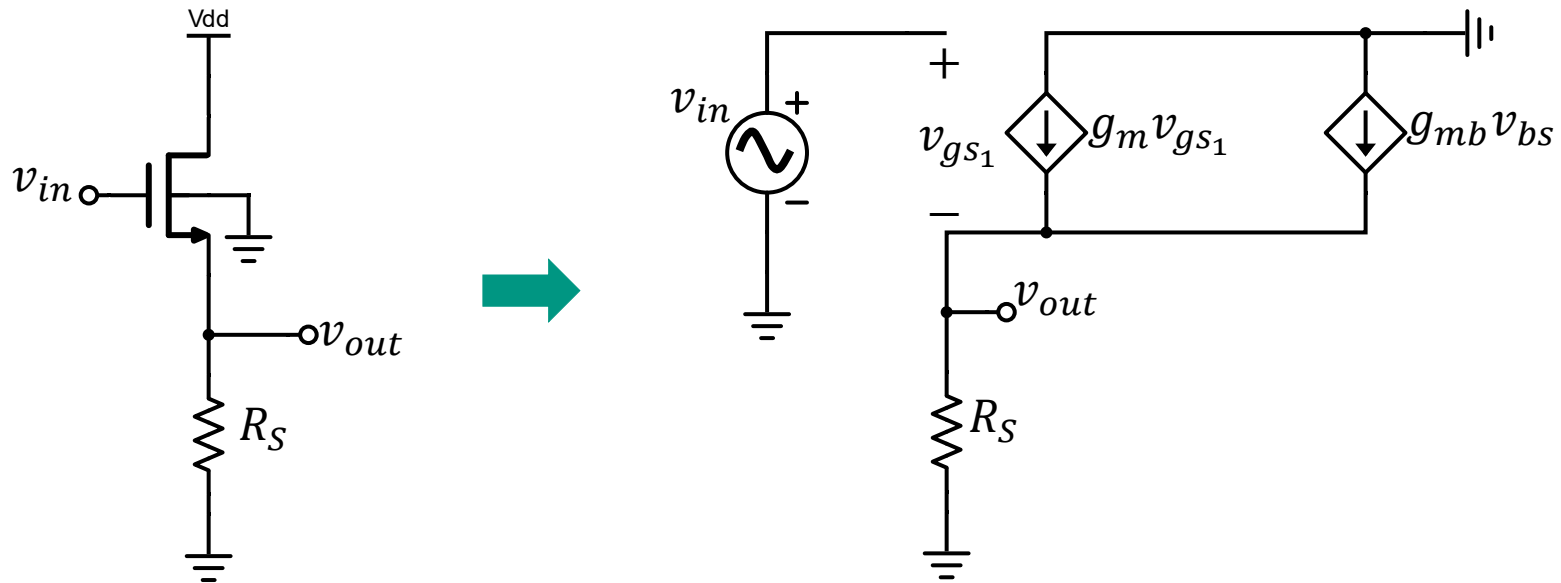


Typical Application



Source Follower (Common-Drain) Stage

Small-Signal Equivalent Circuit

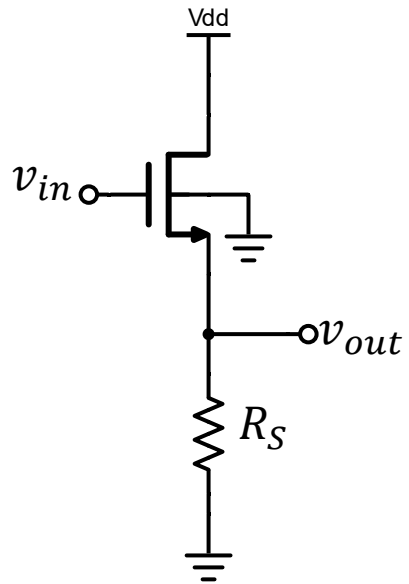


$$g_m v_{gs1} + g_{mb} v_{bs} = \frac{v_{out}}{R_S}, \quad v_{gs1} = v_{in} - v_{out}, \quad v_{bs} = -v_{out}$$

$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_m R_S}{1 + (g_m + g_{mb}) R_S}$$

Source Follower (Common-Drain) Stage

Small-Signal Equivalent



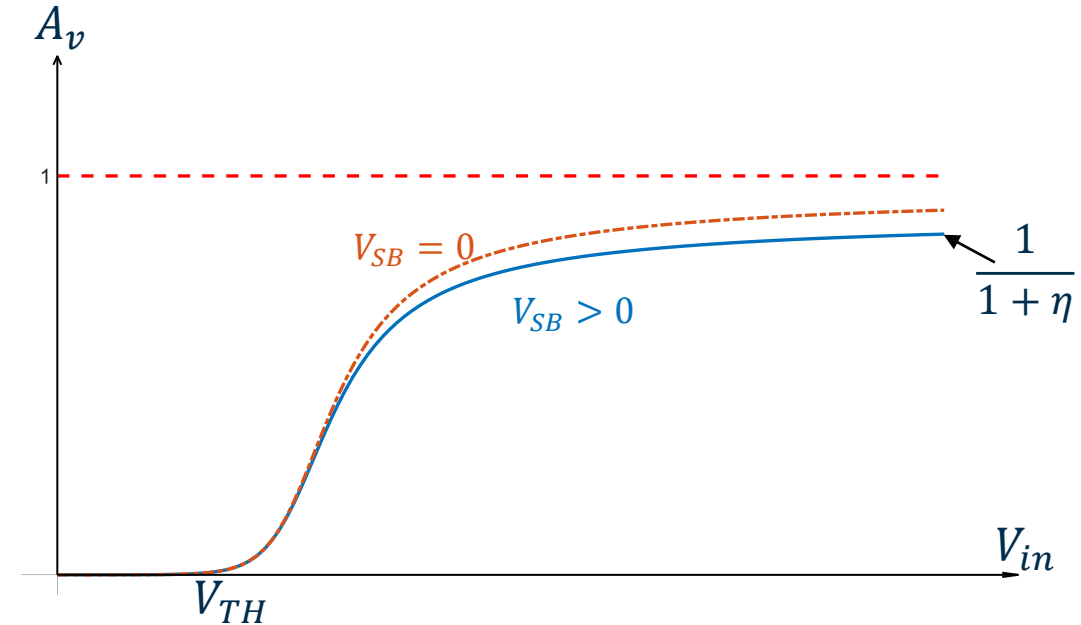
$$A_v = \frac{v_{out}}{v_{in}} = \frac{g_m R_S}{1 + (g_m + g_{mb}) R_S} = \frac{g_m R_S}{1 + (1 + \eta) g_m R_S}$$

- For $g_m R_S \gg 1$,

$$A_v = \frac{v_{out}}{v_{in}} \approx \frac{1}{1 + \eta}$$

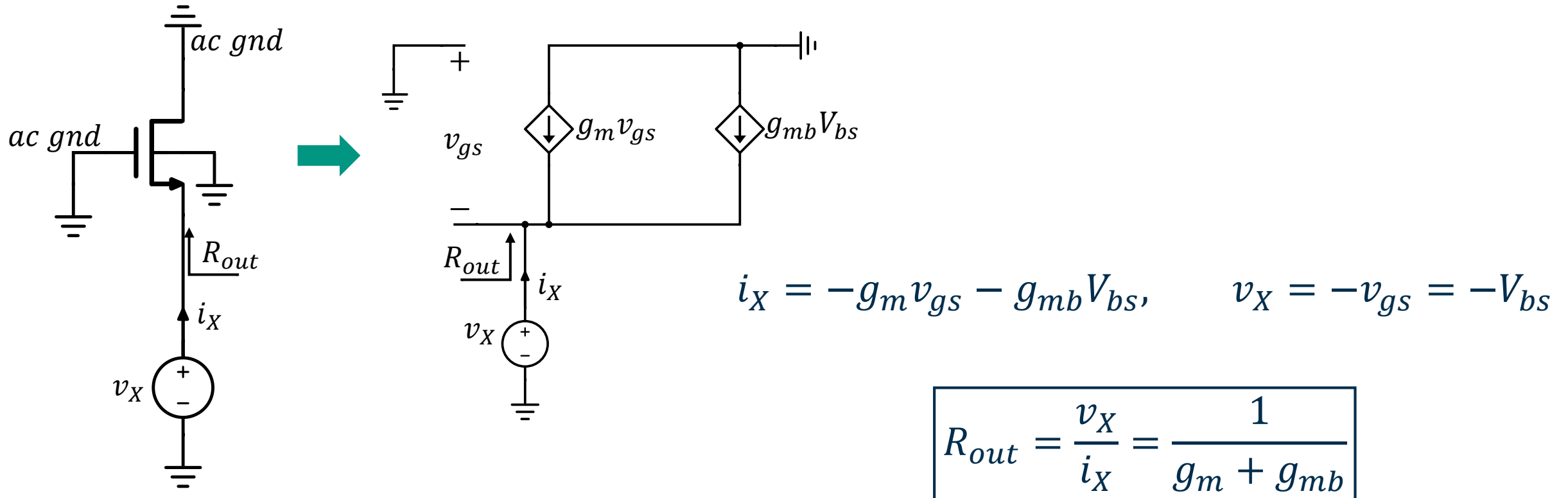
- Ignoring g_{mb} we get

$$A_v = \frac{v_{out}}{v_{in}} \approx 1$$



Source Follower (Common-Drain) Stage

Output Resistance



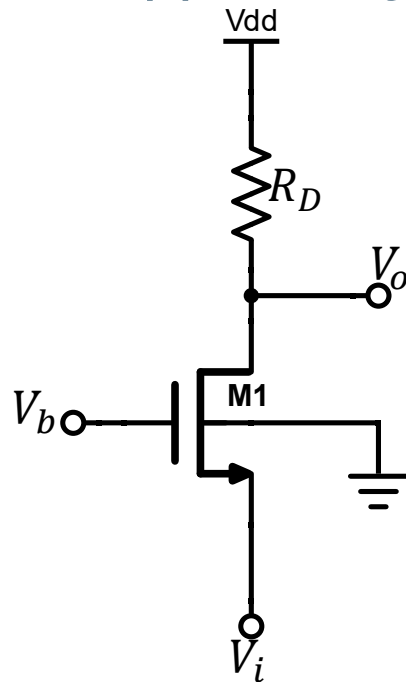


Common-Gate Amplifier

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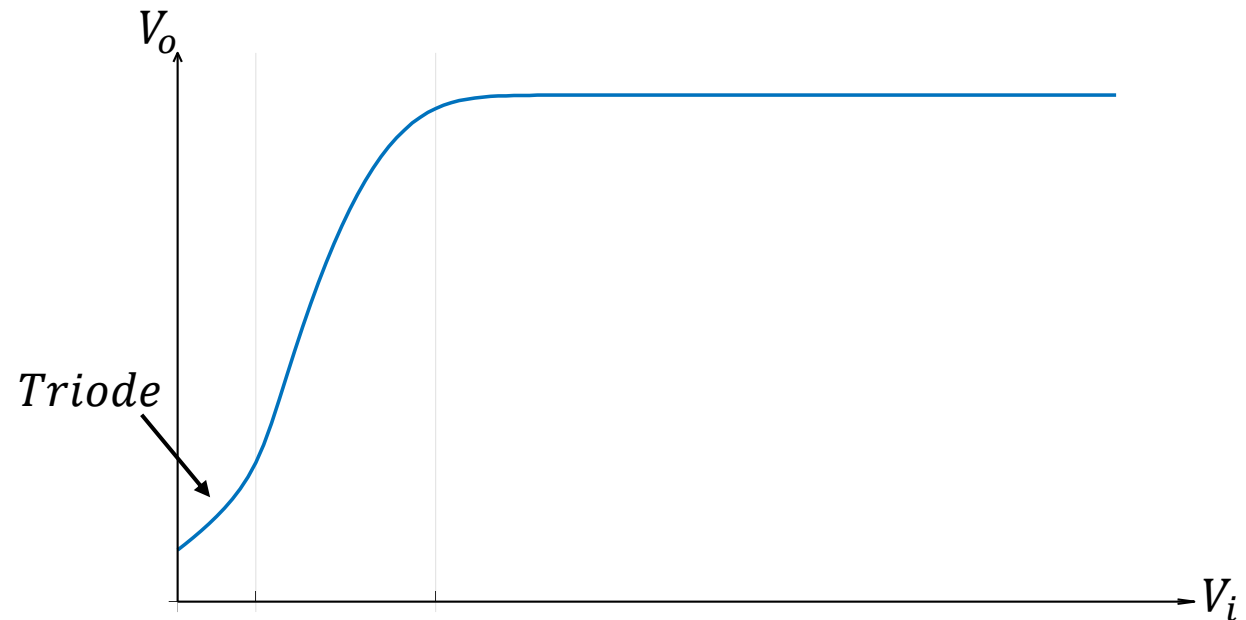
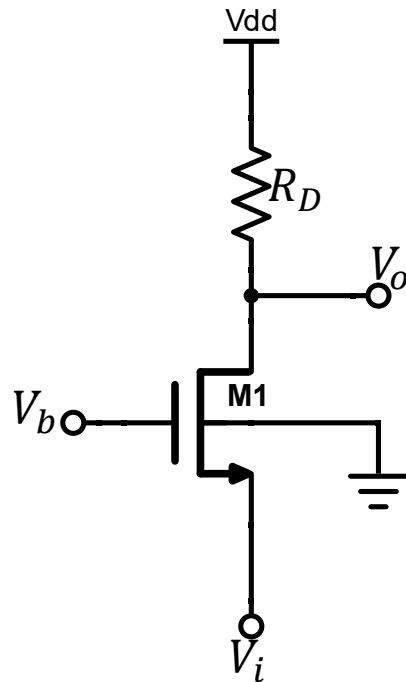
Common-Gate Stage

- The common-gate amplifier provides low input and high output impedance, operates as a non-inverting voltage amplifier. Since it offers approximately unity current gain, it is used as a current buffer. It is widely used in RF applications and forms the upper stage of the cascode amplifier.



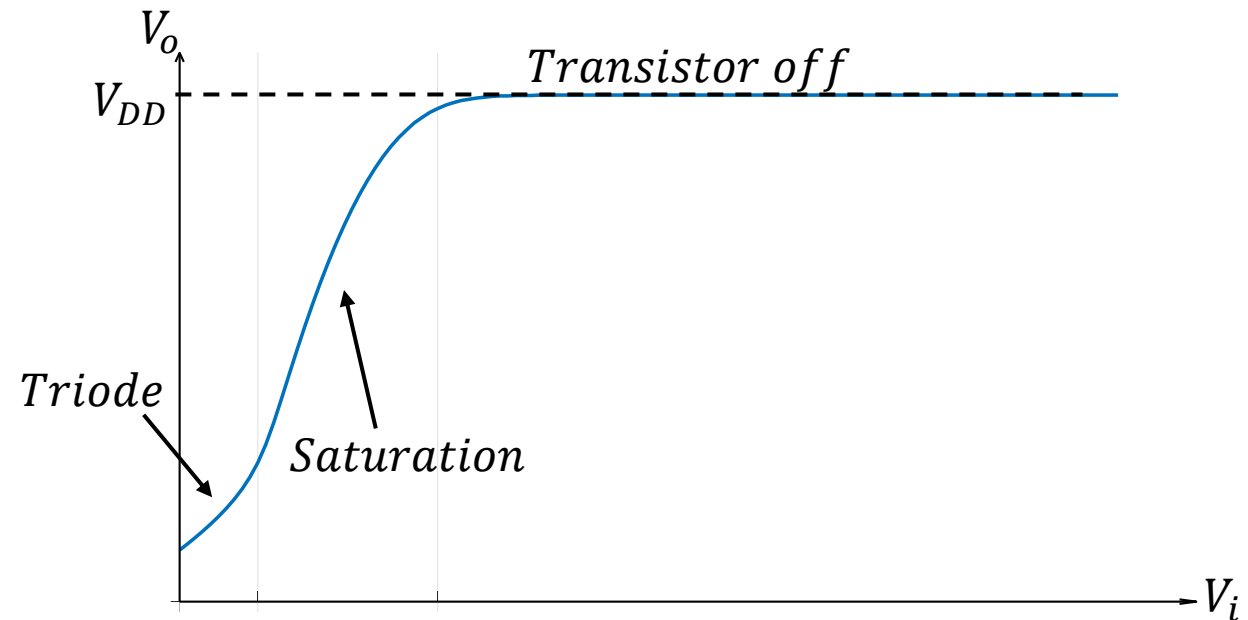
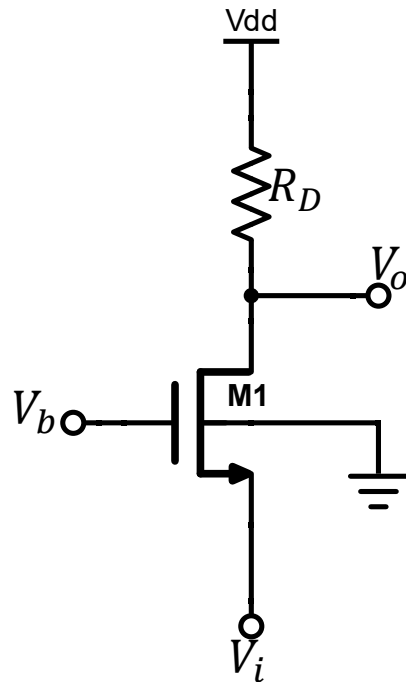
Common-Gate Stage

- At the DC level, V_b is fixed. As V_i increases, the transistor initially exhibits a very high overdrive ($V_{ov} = V_b - V_i - V_{TH}$) voltage, therefore, very low on-resistance (r_{on}), resulting in V_{DS} ($V_o - V_i$) lower than V_{ov} . Therefore, the MOSFET operates first in the **triode region**.



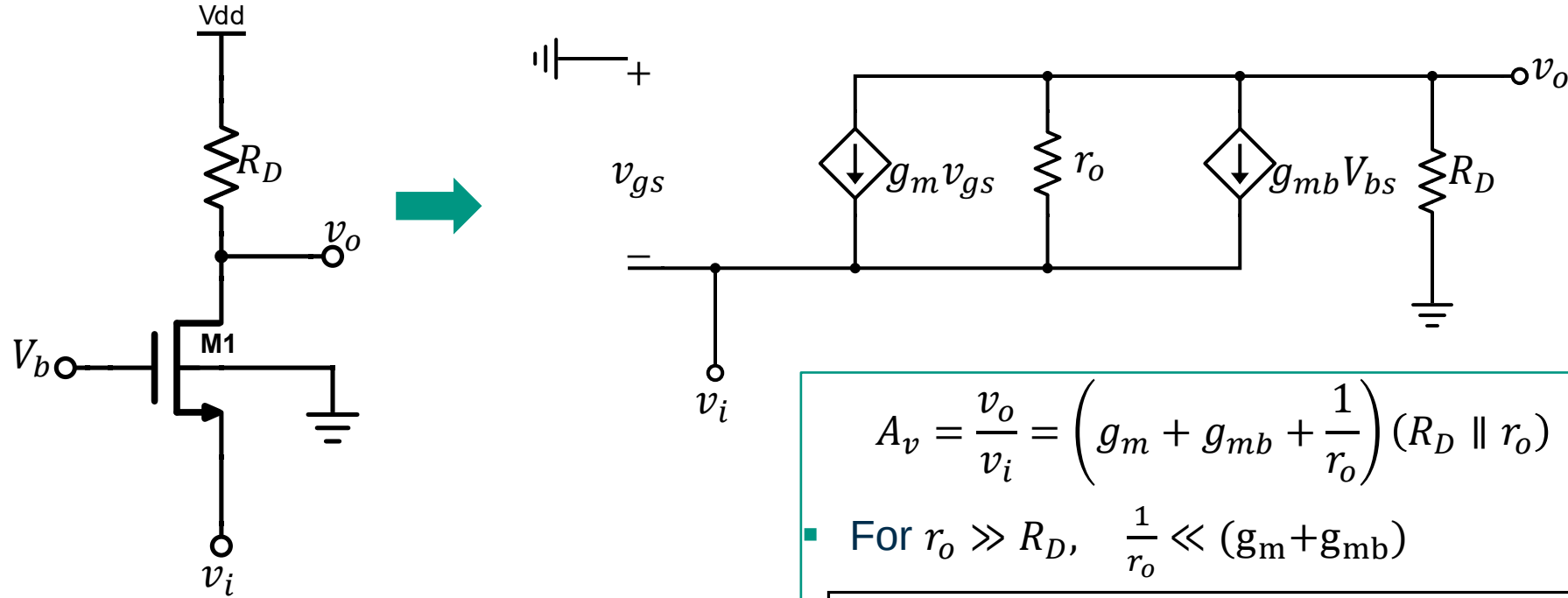
Common-Gate Stage

- As V_i continues to increase, V_{ov} decreases, which increases r_{on} . This will satisfy $V_{DS} > V_{ov}$. → **Saturation**.
- Eventually, as $V_b - V_i$ drops below V_{TH} , **the transistor turns off** and drain voltage approaches to V_{DD} .



Common-Gate Stage

Small Signal Analysis



$$A_v = \frac{v_o}{v_i} = \left(g_m + g_{mb} + \frac{1}{r_o} \right) (R_D \parallel r_o)$$

- For $r_o \gg R_D$, $\frac{1}{r_o} \ll (g_m + g_{mb})$

$$A_v = \frac{v_o}{v_i} \approx (g_m + g_{mb}) R_D = g_m (1 + \eta) R_D$$

- Note that the gain is positive, and body effect increases the equivalent transconductance.

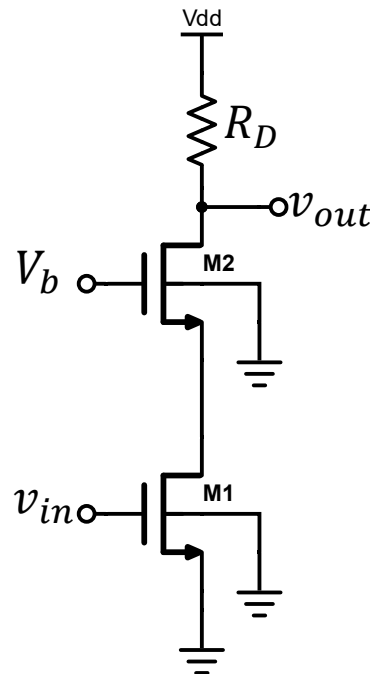


Cascode Amplifier

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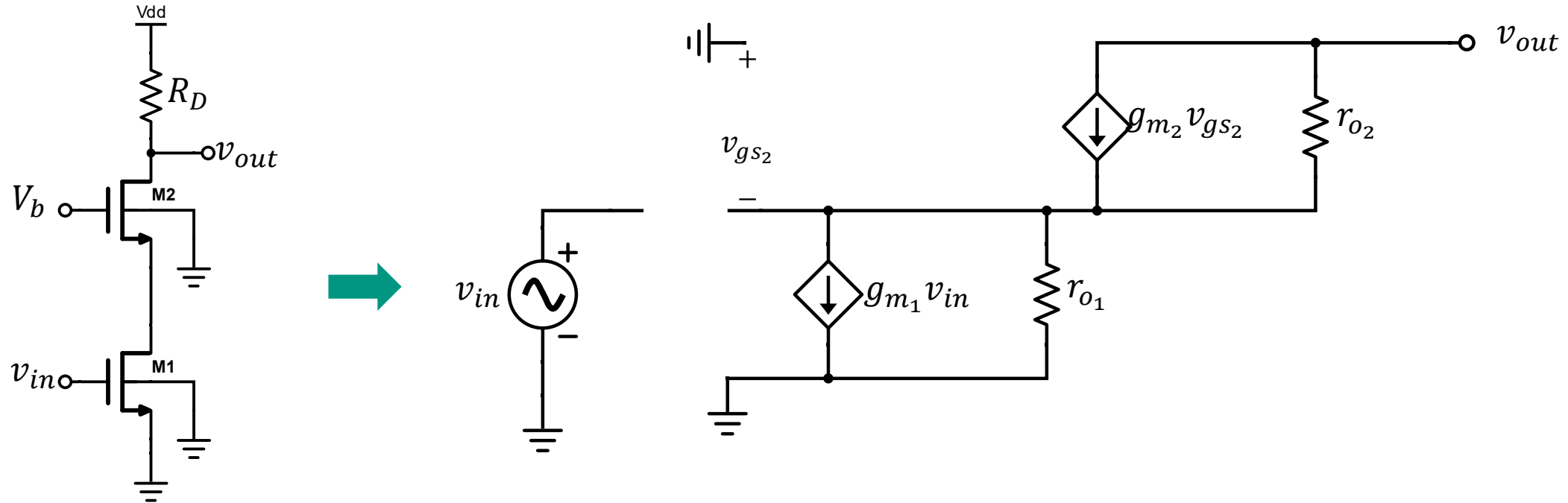
Cascode Stage

- The cascade of a CS stage and a CG stage is called a “cascode”. This topology is providing many useful properties.
- Compared to a single transistor amplifier, it offers a higher output impedance, improved gain bandwidth, and better isolation between the input and output. It has the ability to suppress the Miller effect, thereby enhancing high-frequency performance.



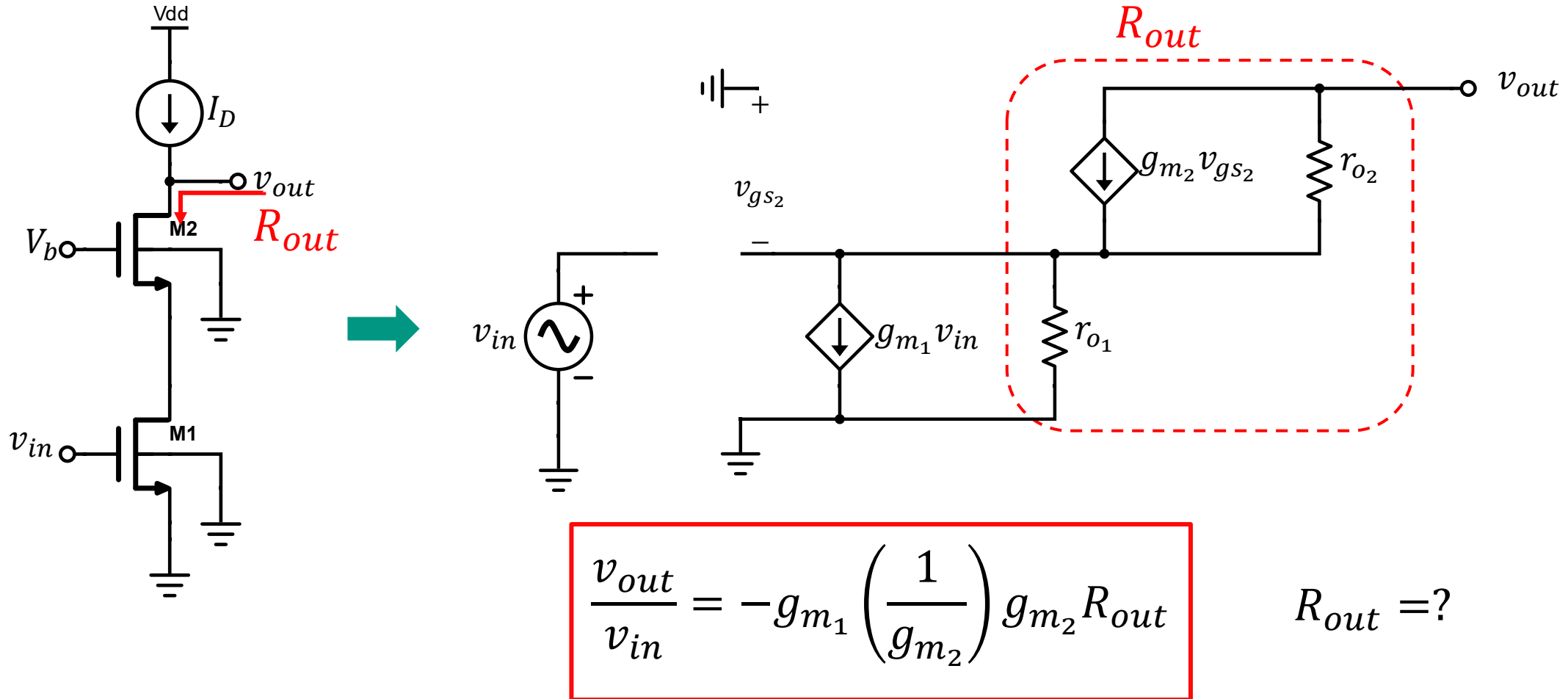
Cascode Stage

Small-Signal Analysis (Body Effect Omitted)



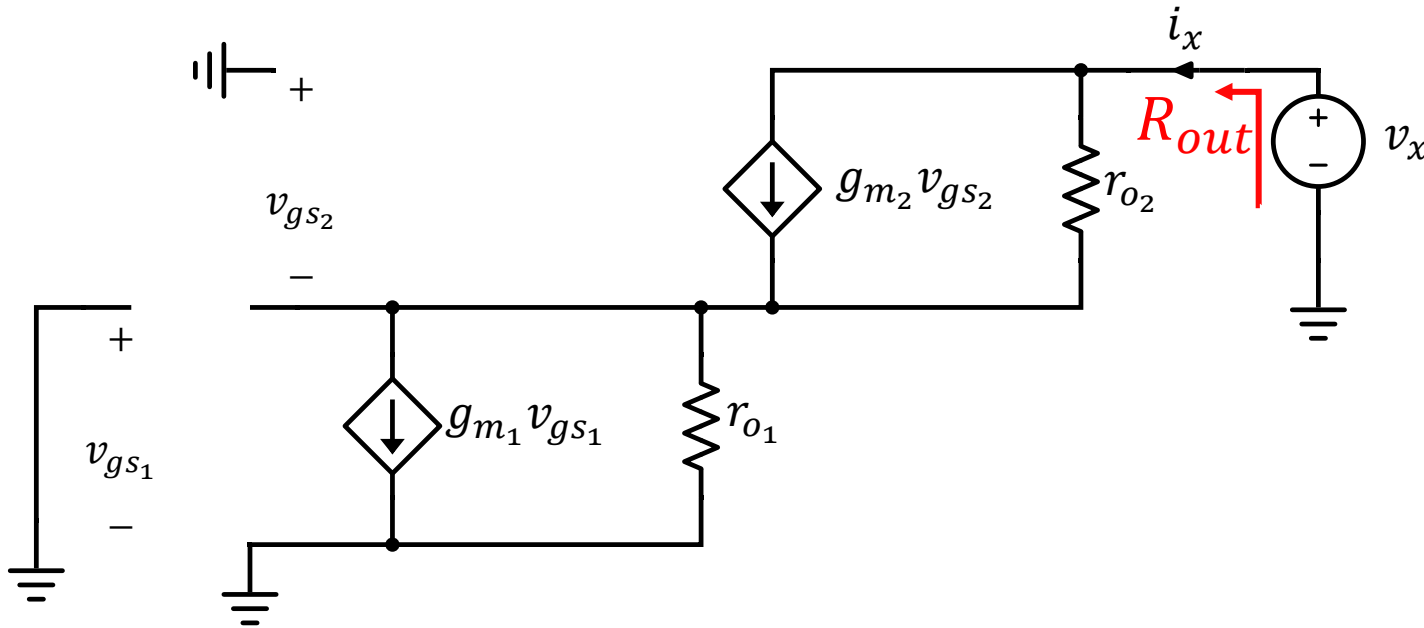
Cascode Stage

Small-Signal Analysis (Body Effect Omitted)



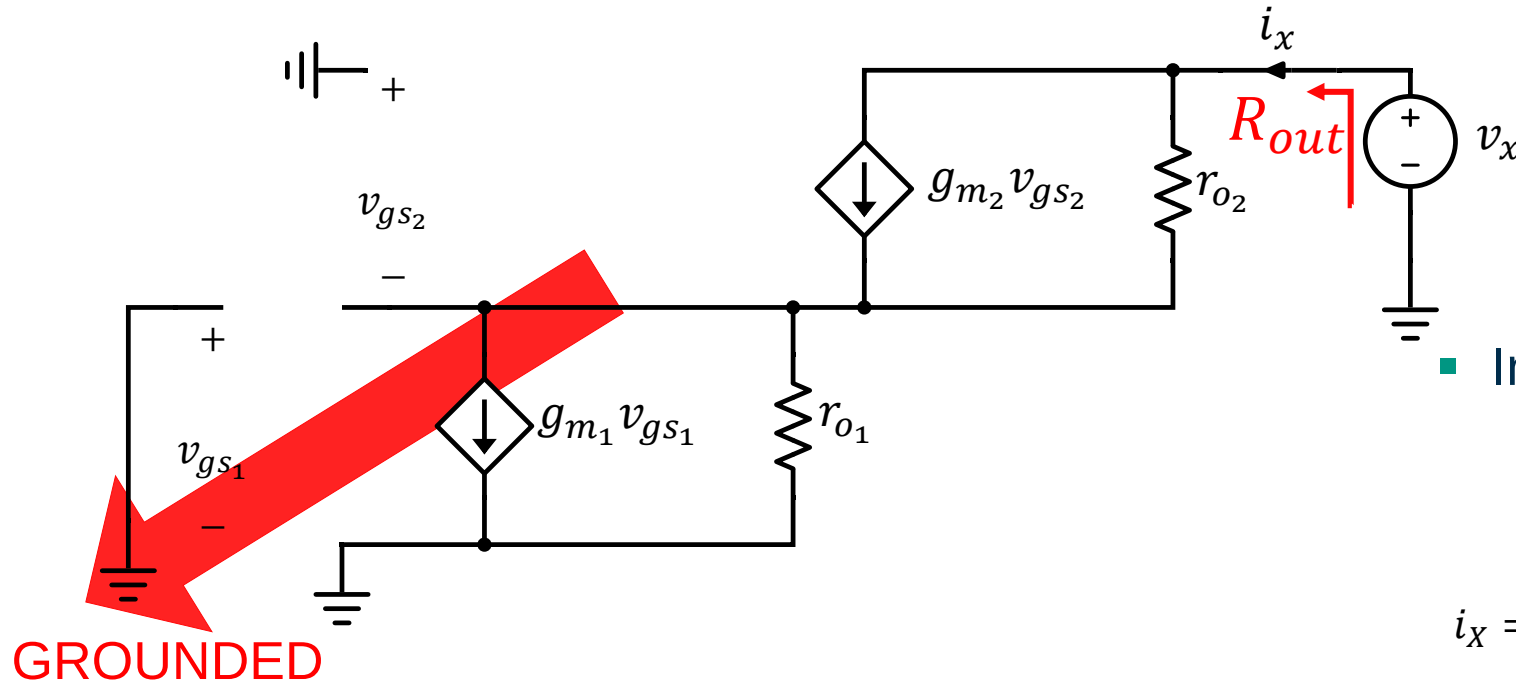
Cascode Stage

Small-Signal Analysis (Body Effect Omitted) – R_{out} Analysis



Cascode Stage

Small-Signal Analysis (Body Effect Omitted) – R_{out} Analysis



- Input port is grounded.

$$v_{gs1} = 0 \Rightarrow g_{m1} v_{gs1} = 0$$

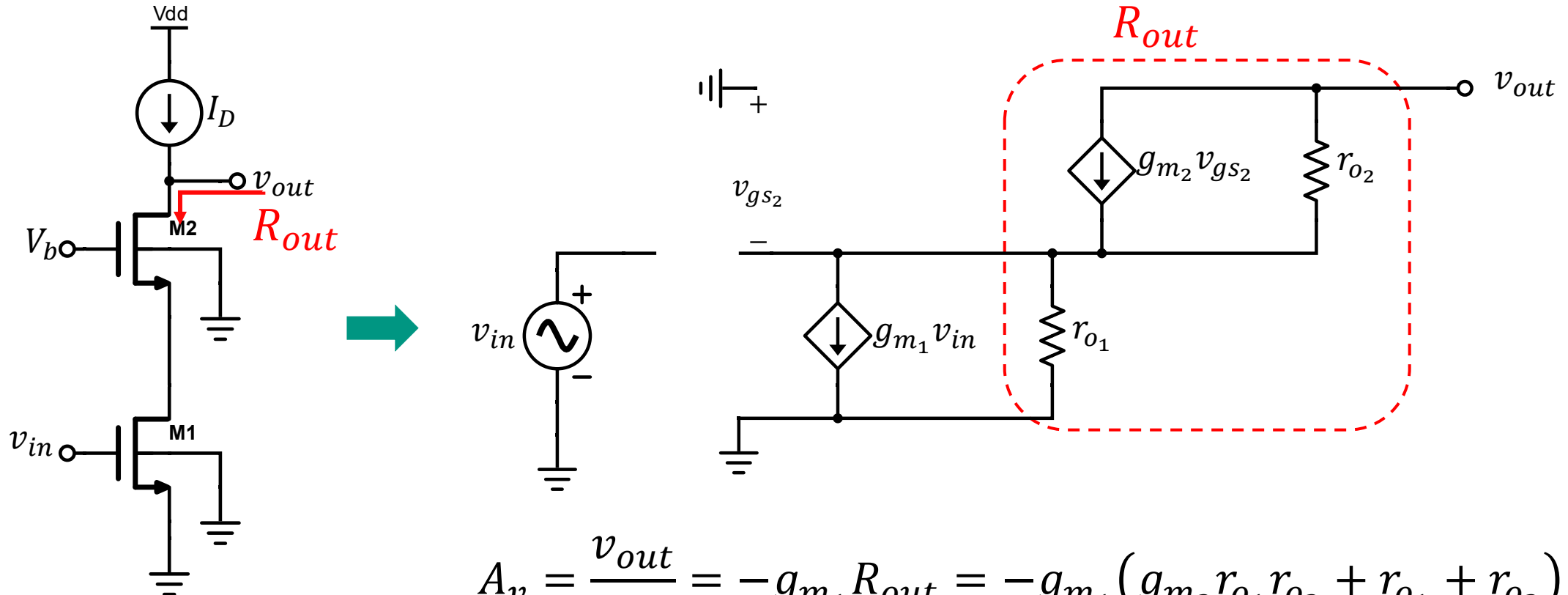
$$v_{gs2} = -i_x r_{o1}$$

$$i_x = g_{m2} v_{gs2} + \frac{(v_x + v_{gs2})}{r_{o2}} = -i_x g_{m2} r_{o1} - i_x \frac{r_{o1}}{r_{o2}} + \frac{v_x}{r_{o2}}$$

$$R_{out} = \frac{v_x}{i_x} = g_{m2} r_{o1} r_{o2} + r_{o1} + r_{o2}$$

Cascode Stage

Small-Signal Analysis (Body Effect Omitted)

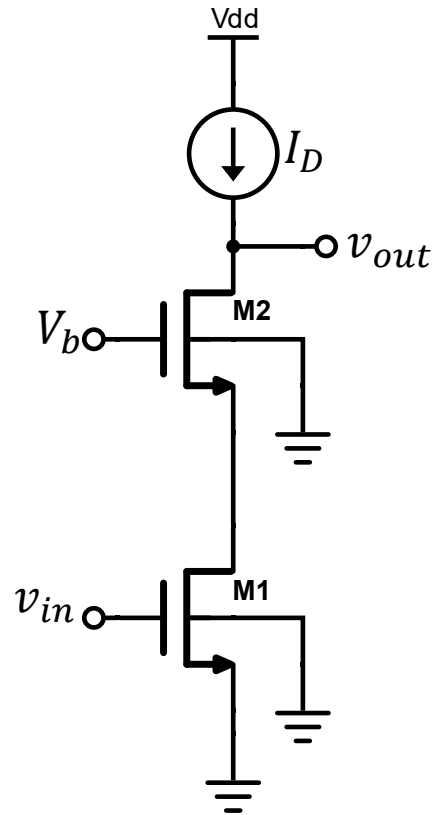


$$A_v = \frac{v_{out}}{v_{in}} = -g_{m1}R_{out} = -g_{m1}(g_{m2}r_{o1}r_{o2} + r_{o1} + r_{o2})$$

$$A_v \approx -g_{m1}g_{m2}r_{o1}r_{o2}$$

Cascode Stage

Small-Signal Analysis (Body Effect Omitted)

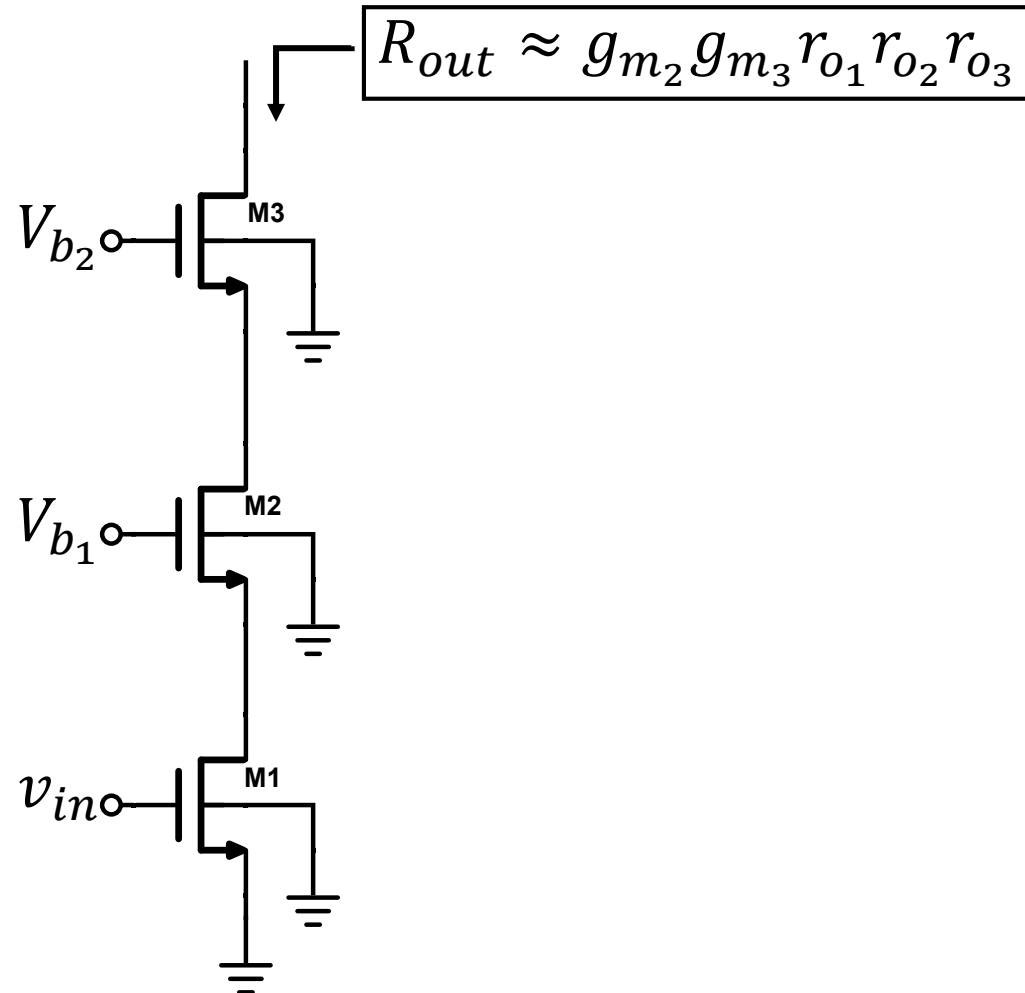


$$A_v \approx -g_{m_1} g_{m_2} r_{o_1} r_{o_2}$$

- The output resistance is boosted by a factor of the intrinsic gain of the cascading transistor, $g_{m_2} r_{o_2}$.
- The boosted resistance multiplies the overall gain.
- Note that for $g_{m_1} = g_{m_2}$ and $r_{o_1} = r_{o_2}$, the overall gain simplifies to $|A_v| = (g_m r_o)^2$.

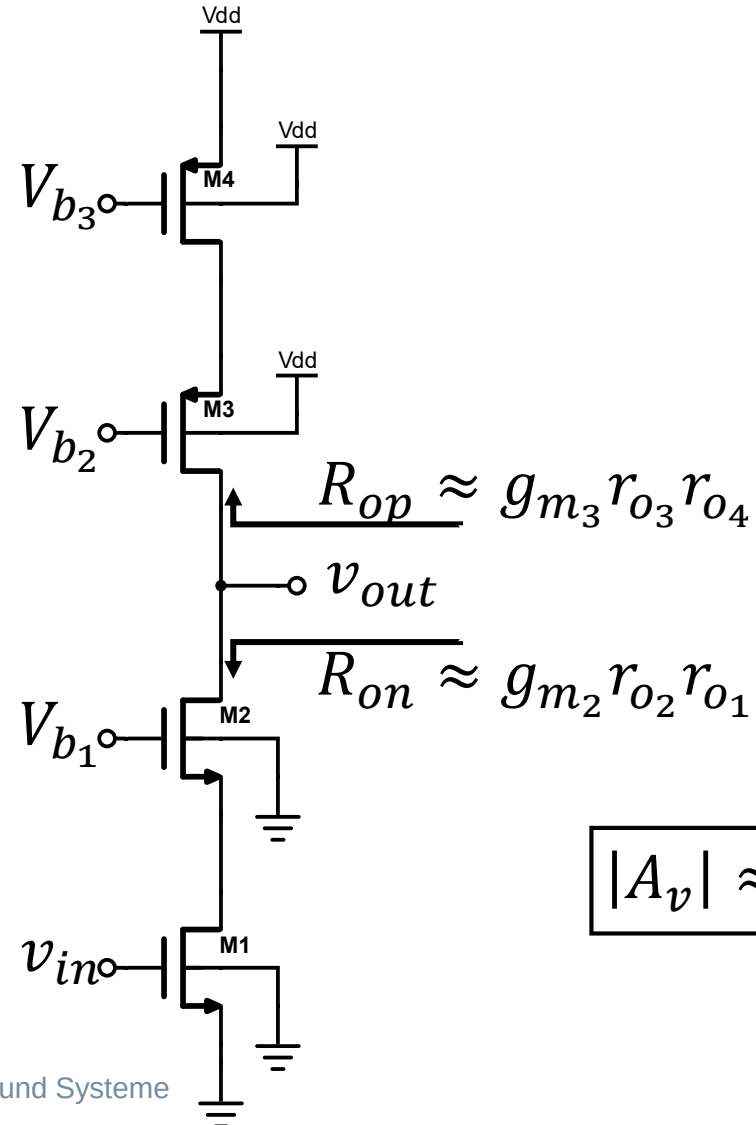
Cascode Stage

Triple Cascode



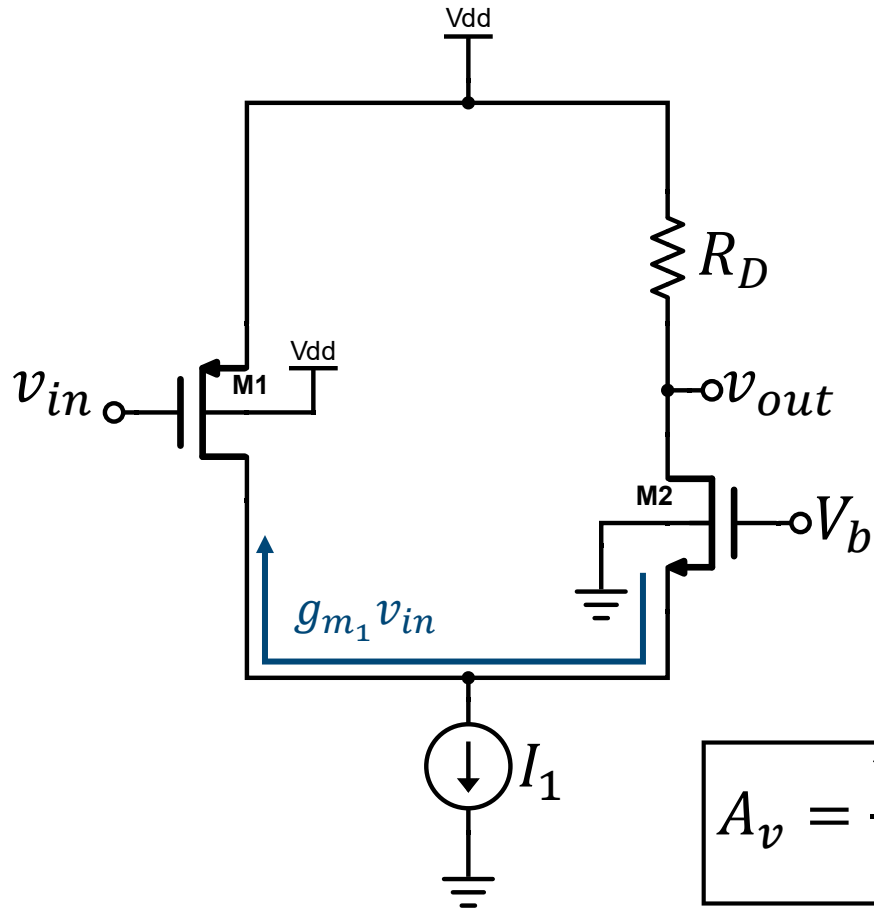
Cascode Stage

NMOS Cascode Amplifier with PMOS Cascode Load



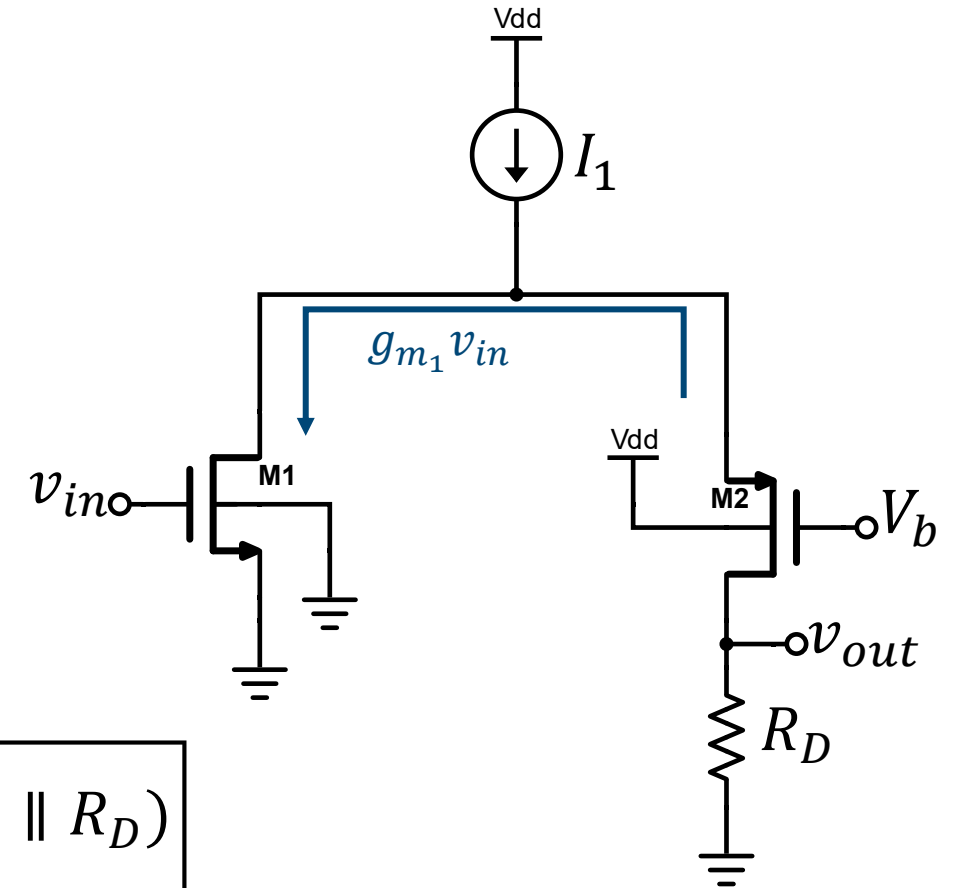
$$|A_v| \approx g_{m1} (R_{op} \parallel R_{on})$$

Folded Cascode



$$A_v = \frac{v_{out}}{v_{in}} \approx -g_{m_1} (g_{m_2} r_{o_2} r_{o_1} \parallel R_D)$$

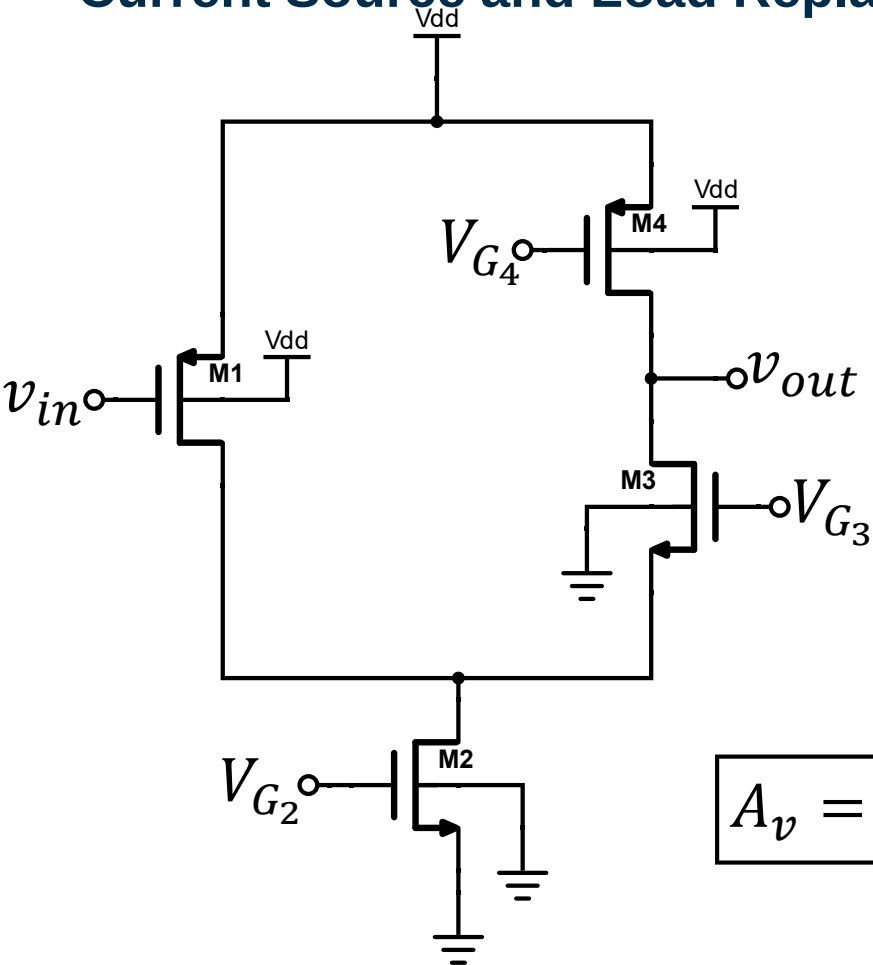
- With PMOS input



- With NMOS input

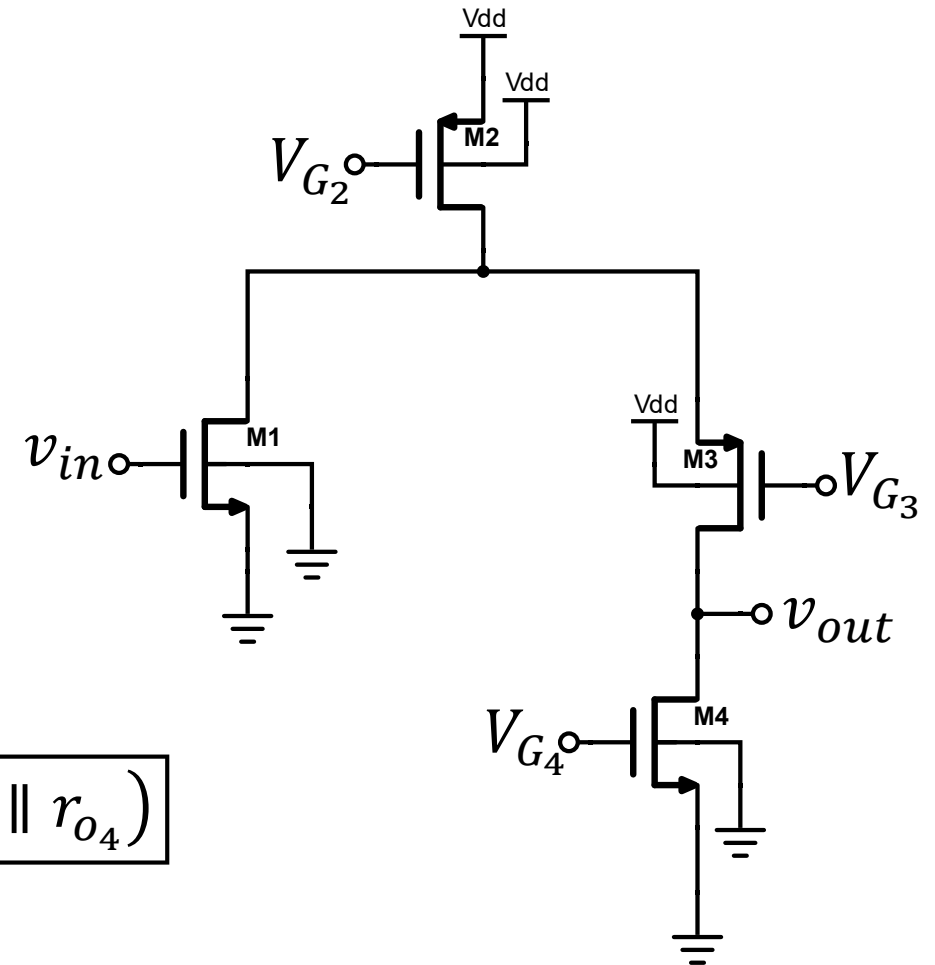
Folded Cascode

Current Source and Load Replaced with Active Devices



$$A_v = -g_{m_1} \left(\left[g_{m_3} r_{o_3} (r_{o_1} \parallel r_{o_2}) \right] \parallel r_{o_4} \right)$$

- With PMOS input



- With NMOS input